Software Requirements Specification

for

Smart Hospital Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The system manages hospital activities such as patient registration, appointments, medical records, billing, lab tests, prescriptions, medicines, and staff management. This SRS covers the overall hospital management system and its major user roles.

1.2 Document Conventions

This SRS uses Arial/Times New Roman, with bold headings for sections and subsections. Requirements are written as clear and numbered statements. Each requirement is assigned its own priority (High, Medium, or Low).

1.3 Intended Audience and Reading Suggestions

*This SRS is intended for **developers, project managers, testers, and end users**. Developers use it for implementation, testers for verification, and project managers/users for understanding the system.*

*The document is organized into **introduction, overall description, functional requirements, non-functional requirements, and system features**. Readers should first read the introduction and overview, then proceed to the sections relevant to their role.*

1.4 Product Scope

The Smart Hospital Management System (SHMS) is designed to digitize and streamline hospital operations. It helps manage patients, appointments, medical records, prescriptions, billing, lab tests, and staff efficiently. The main goals are to reduce paperwork, improve accuracy, save time, and provide better patient care through a centralized and user-friendly system.

1.5 References

No external documents or web references are used for this SRS. The requirements are based on the Smart Hospital Management System project specifications and use-case documentation prepared by the project team.

2. Overall Description

2.1 Product Perspective

The Smart Hospital Management System (SHMS) is a new, self-contained software product developed to digitize and simplify hospital operations. It provides a centralized platform for managing patients, doctors, appointments, medical records, laboratory tests, medicines, and billing. The system connects different hospital users such as patients, doctors, receptionists, nurses, pharmacists, and administrators through a common database and user interface.

2.2 Product Functions

The Smart Hospital Management System provides the following major functions:

- *Login and user authentication.*
- *Patient registration and profile management.*
- *Appointment booking, viewing, and cancellation.*
- *Medical record management and patient diagnosis.*
- *Prescription generation and management.*
- *Laboratory test request, conducting, and result updating.*
- *Bill generation and payment processing.*
- *Doctor and staff management.*
- *Medicine management.*
- *Hospital report generation.*

2.3 User Classes and Characteristics

The Smart Hospital Management System is intended for the following user classes:

- *Patient – Registers with the system, books appointments, views medical records, and makes payments. Basic computer knowledge is sufficient.*
- *Doctor – Views patient records, diagnoses patients, provides prescriptions, and requests laboratory tests. Requires medical knowledge and basic computer skills.*
- *Admin – Manages doctors, staff, patients, and system information. Has the highest level of access and requires good technical knowledge.*
- *Receptionist – Registers patients, manages appointments, and handles billing. Requires basic computer skills.*
- *Nurse – Views and updates patient information and assists doctors in patient care. Requires basic system knowledge.*
- *Laboratory Staff – Manages laboratory tests and updates test results. Requires knowledge of laboratory procedures and basic computer skills.*
- *Pharmacist – Manages medicines and prescriptions. Requires basic computer skills and knowledge of medicines.*

2.4 Operating Environment

The Smart Hospital Management System will operate in the following environment:

- *Hardware: Desktop computers, laptops, tablets, and smartphones with internet connectivity.*
- *Operating System: Windows 10/11, macOS, Linux, Android, and iOS.*
- *Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.*
- *Server: A web server capable of running the application and database.*
- *Database: MySQL or MongoDB.*
- *Network: Stable internet connection or hospital local area network (LAN).*
- *Other Software: The system should coexist with standard web browsers, office applications, and other hospital management software.*

2.5 Design and Implementation Constraints

The development of the Smart Hospital Management System will be subject to the following constraints:

- *The system must ensure the security and privacy of patient information.*
- *The system must use a reliable database for storing hospital data.*
- *The system must support secure user authentication and role-based access.*
- *The system must be accessible through standard web browsers.*
- *The system should provide reliable performance with multiple users accessing it simultaneously.*
- *The system should follow standard programming, database, and security practices.*
- *The system requires a stable network connection for proper operation.*

2.6 User Documentation

<List the user documentation components (such as user manuals, on-line help, and tutorials) that will be delivered along with the software. Identify any known user documentation delivery formats or standards.>

2.7 Assumptions and Dependencies

<List any assumed factors (as opposed to known facts) that could affect the requirements stated in the SRS. These could include third-party or commercial components that you plan to use, issues around the development or operating environment, or constraints. The project could be affected if these assumptions are incorrect, are not shared, or change. Also identify any dependencies the project has on external factors, such as software components that you intend to reuse from another project, unless they are already documented elsewhere (for example, in the vision and scope document or the project plan).>

3. External Interface Requirements

3.1 User Interfaces

<Describe the logical characteristics of each interface between the software product and the users. This may include sample screen images, any GUI standards or product family style guides that are to be followed, screen layout constraints, standard buttons and functions (e.g., help) that will appear on every screen, keyboard shortcuts, error message display standards, and so on. Define the software components for which a user interface is needed. Details of the user interface design should be documented in a separate user interface specification.>

3.2 Hardware Interfaces

<Describe the logical and physical characteristics of each interface between the software product and the hardware components of the system. This may include the supported device types, the nature of the data and control interactions between the software and the hardware, and communication protocols to be used.>

3.3 Software Interfaces

<Describe the connections between this product and other specific software components (name and version), including databases, operating systems, tools, libraries, and integrated commercial components. Identify the data items or messages coming into the system and going out and describe the purpose of each. Describe the services needed and the nature of communications. Refer to documents that describe detailed application programming interface protocols. Identify data that will be shared across software components. If the data sharing mechanism must be implemented in a specific way (for example, use of a global data area in a multitasking operating system), specify this as an implementation constraint.>

3.4 Communications Interfaces

<Describe the requirements associated with any communications functions required by this product, including e-mail, web browser, network server communications protocols, electronic forms, and so on. Define any pertinent message formatting. Identify any communication standards that will be used, such as FTP or HTTP. Specify any communication security or encryption issues, data transfer rates, and synchronization mechanisms.>

4. System Features

<This template illustrates organizing the functional requirements for the product by system features, the major services provided by the product. You may prefer to organize this section by use case, mode of operation, user class, object class, functional hierarchy, or combinations of these, whatever makes the most logical sense for your product.>

4.1 System Feature 1

4.1.1 Description and Priority

The Patient Management feature allows authorized users to register patients, maintain patient profiles, and view and update patient information. **Priority: High.**

4.1.2 Stimulus/Response Sequences

- User enters patient details → System validates and stores the information.
- User searches for a patient → System displays the patient's information.
- User updates patient details → System validates and updates the information.
- User enters invalid or incomplete data → System displays an appropriate error message.

4.1.3 Functional Requirements

REQ-1: The system shall allow authorized users to register new patients.

REQ-2: The system shall store the patient's personal and contact information.

REQ-3: The system shall generate a unique patient ID for each patient.

REQ-4: The system shall allow authorized users to search and view patient information.

REQ-5: The system shall allow authorized users to update patient information.

REQ-6: The system shall validate required fields before saving patient information.

REQ-7: The system shall display an error message when invalid or incomplete information is entered.

Appointment Management

4.2.1 Description and Priority

The Appointment Management feature allows patients and authorized staff to book, view, and cancel appointments with doctors. **Priority: High.**

4.2.2 Stimulus/Response Sequences

- Patient selects a doctor and available time → System confirms the appointment.
- User views appointments → System displays the appointment details.
- User cancels an appointment → System updates the appointment status.

- User selects an unavailable time → System displays an appropriate error message.

4.2.3 Functional Requirements

REQ-8: The system shall allow patients to book appointments with doctors.

REQ-9: The system shall display available appointment slots.

REQ-10: The system shall store appointment details.

REQ-11: The system shall allow authorized users to view appointments.

REQ-12: The system shall allow patients to cancel appointments.

REQ-13: The system shall prevent booking of unavailable appointment slots.

REQ-14: The system shall display an error message for invalid appointment requests.

5. Other Nonfunctional Requirements

Performance Requirements

- *The system shall respond to normal user requests within **3 seconds** under normal operating conditions.*
- *The system shall support multiple users accessing the system simultaneously.*
- *The system shall process appointment booking and cancellation requests without unnecessary delay.*
- *The system shall maintain consistent performance during normal hospital operations.*

Safety Requirements

- *The system shall prevent unauthorized modification or deletion of important patient information.*
- *The system shall provide confirmation before critical actions such as deleting records or cancelling appointments.*
- *The system shall maintain accurate patient and medical information to reduce the risk of incorrect treatment decisions.*
- *The system shall provide regular data backup to prevent loss of important information.*

Security Requirements

- *The system shall require users to log in using valid credentials.*
- *The system shall provide **role-based access control** for patients, doctors, nurses, administrators, receptionists, laboratory staff, and pharmacists.*
- *The system shall prevent users from accessing information outside their assigned permissions.*
- *Patient and medical information shall be protected from unauthorized access.*
- *The system shall securely store user credentials and sensitive data.*
- *The system shall automatically terminate inactive user sessions after a defined period.*

Software Quality Attributes

- ***Usability:** The system shall provide a simple and user-friendly interface.*
- ***Reliability:** The system shall operate reliably and minimize unexpected failures.*
- ***Availability:** The system should be available whenever hospital services are in operation.*
- ***Maintainability:** The system shall be designed so that updates and maintenance can be performed easily.*
- ***Performance:** The system shall provide responses within an acceptable time under normal usage.*
- ***Security:** The system shall protect patient and hospital information from unauthorized access.*
- ***Scalability:** The system should support an increasing number of patients, staff, and records.*

Business Rules

- *Only authorized users shall access hospital information according to their roles.*
- *Patients can book and cancel their own appointments.*
- *Doctors can view patient medical records and provide diagnoses and prescriptions.*
- *Nurses can access and update information relevant to patient care.*
- *Receptionists can register patients, manage appointments, and handle billing.*
- *Laboratory staff can manage laboratory tests and update test results.*
- *Pharmacists can manage medicines and process prescriptions.*
- *Administrators have permission to manage users, staff, and system information.*

6. Other Requirements

<Define any other requirements not covered elsewhere in the SRS. This might include database requirements, internationalization requirements, legal requirements, reuse objectives for the project, and so on. Add any new sections that are pertinent to the project.>

Appendix A: Glossary

<Define all the terms necessary to properly interpret the SRS, including acronyms and abbreviations. You may wish to build a separate glossary that spans multiple projects or the entire organization, and just include terms specific to a single project in each SRS.>

Appendix B: Analysis Models

<Optionally, include any pertinent analysis models, such as data flow diagrams, class diagrams, state-transition diagrams, or entity-relationship diagrams.>

Appendix C: To Be Determined List

<Collect a numbered list of the TBD (to be determined) references that remain in the SRS so they can be tracked to closure.>